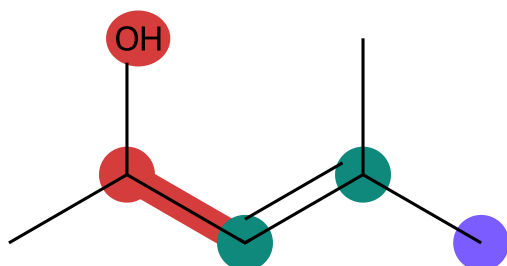


PART A · BUILDING A NAME

1 Anatomy of a Name



4-methylpent-3-en-2-ol

■ prefix	substituent	<i>4-methyl</i>
■ root	chain length	<i>pent · 5 C</i>
■ 1° suffix	saturation	<i>-3-ene</i>
■ 2° suffix	main group	<i>-2-ol</i>

Assembled in one fixed order — locants pin each piece to a carbon:

(**prefixes**)(**locants**)-(**root**)(**1° suffix**)(**2° suffix**)

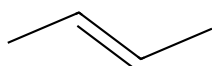
2 Roots & Primary Suffixes

Root = carbon count · primary suffix = degree of saturation

C	ROOT
1	meth
2	eth
3	prop
4	but
5	pent
6	hex

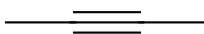
C	ROOT
7	hept
8	oct
9	non
10	dec
11	undec
12	dodec

- All single bonds → **-ane** · C=C → **-ene** · C≡C → **-yne**
- Several such bonds → **-diene** / **-triene** / **-diyne**; keep an **a** before a consonant suffix: penta**a**-1,3-diene



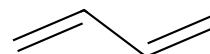
but-2-ene

C=C at 2



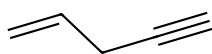
but-2-yne

C≡C at 2



buta-1,3-diene

two C=C



pent-1-en-4-yne

-en- before -yne

3 Selecting the Parent Chain

Apply in order; move on only to break a tie

- Most of the **principal characteristic group** (the senior group)
- Most skeletal **multiple bonds** (C=C and C≡C counted together)
- **Longest** chain, then most substituents cited as prefixes

The principal group **must lie in the parent chain** and alone takes the suffix. The longest chain is often **not** the one drawn horizontally — trace every path.

4 Numbering the Chain

Lowest locants by first point of difference

- Lowest to the **principal group** (the suffix)
- Then lowest to **multiple bonds** (= and ≡ together; C=C wins a tie)
- Then lowest **set** to the prefixes
- Then lowest to the prefix cited **first alphabetically**

First point of difference: compare the two locant sets term by term — the set smaller at the first difference wins: {2,3,5} < {2,4,5}.

5 Functional-Group Seniority

Highest wins the suffix; everything else becomes a prefix

#	CLASS	GROUP
1	Carboxylic acid	—COOH
2	Sulphonic acid	—SO ₃ H
3	Anhydride	—CO—O—CO—
4	Ester	—COOR
5	Acid halide	—COX
6	Amide	—CONH ₂
7	Nitrile	—CN
8	Aldehyde	—CHO
9	Ketone	>C=O
10	Alcohol / Phenol	—OH

11 Amine

 —NH_2

Below amine: ethers, halides, nitro, nitroso, and $\text{C}=\text{C}/\text{C}\equiv\text{C}$ — but **ethers, halides and nitro are always prefixes**. Only a multiple bond among these shows in the suffix (-ene / -yne).

6 Suffixes & Prefixes of Groups

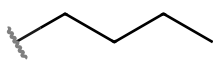
GROUP	SUFFIX (SENIOR)	PREFIX (JUNIOR)
—COOH	-oic acid / -carboxylic acid	carboxy
$\text{—SO}_3\text{H}$	-sulphonic acid	sulpho
—COOR	-oate	alkoxycarbonyl
—COX	-oyl halide	halocarbonyl
—CONH_2	-amide / -carboxamide	carbamoyl
—CN	-nitrile / -carbonitrile	cyano
—CHO	-al / -carbaldehyde	oxo / formyl
$>\text{C}=\text{O}$	-one	oxo
—OH	-ol	hydroxy
—NH_2	-amine	amino
—OR	—	alkoxy
—X	—	halo (fluoro/chloro/bromo/iodo)
—NO_2	—	nitro

Carbon-counting trap: for —COOH , —CHO , —CN , —CONH_2 the group carbon **is counted** in the root (CH_3CN = ethanenitrile, 2 C). The -carbo... ring forms (-carboxylic acid / -carbaldehyde / -carbonitrile) do **not** count it.

7 Substituent (Alkyl) Prefixes

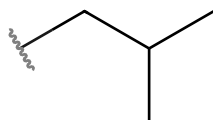
GROUP	COMMON	IUPAC
$\text{CH}_3\text{—}$	methyl	methyl
$\text{C}_2\text{H}_5\text{—}$	ethyl	ethyl
$(\text{CH}_3)_2\text{CH—}$	isopropyl	propan-2-yl
$\text{CH}_2=\text{CH—}$	vinyl	ethenyl
$\text{CH}_2=\text{CHCH}_2\text{—}$	allyl	prop-2-en-1-yl
$\text{C}_6\text{H}_5\text{—}$	phenyl	phenyl
$\text{C}_6\text{H}_5\text{CH}_2\text{—}$	benzyl	phenylmethyl

The four confusable $\text{C}_4\text{H}_9\text{—}$ groups (ν = point of attachment):



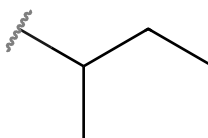
butan-1-yl

n-butyl



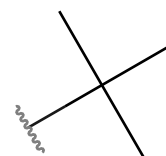
2-methylpropyl

isobutyl



butan-2-yl

sec-butyl



2-methylpropan-2-yl

tert-butyl

8 Multiplying Prefixes

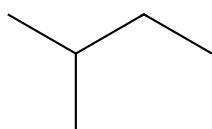
NUMBER	SIMPLE	COMPLEX
2	di	bis
3	tri	tris
4	tetra	tetrakis
5	penta	pentakis

Use **bis** / **tris** / **tetrakis** for complex (substituted) substituents — bis(2-chloroethyl). Multiplying prefixes are **ignored when alphabetising**.

PART B · NAMING BY FUNCTIONAL CLASS

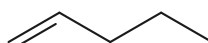
9 Alkanes · Alkenes · Alkynes

- Saturated → **-ane**; $C=C$ → **-ene**; $C\equiv C$ → **-yne**, each with its lowest locant
- Both present → **-en...-yne**; on a tie the $C=C$ takes the lower number → but-1-en-3-yne



2-methylbutane

branched alkane



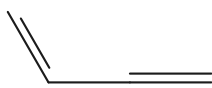
pent-1-ene

$C=C$ at 1



pent-1-yne

$C\equiv C$ at 1

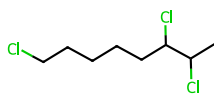


but-1-en-3-yne

$C=C$ wins the tie

10 Lowest Set of Locants

Lowest at the first difference — not the smallest total



1,6,7-trichlorooctane

$\{1, 6, 7\}$ beats $\{2, 3, 8\}$

From the left → **{1, 6, 7}** ✓ (sum 14)

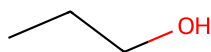
From the right → **{2, 3, 8}** ✗ (sum 13)

1 < 2 at the very first term wins — the smaller *total* is irrelevant.

Applied to the **whole set at once**, in priority order: principal group > multiple bonds > all prefixes together > first-alphabetical prefix. Sort each candidate set ascending and compare position by position.

11 Alcohols & Ethers

- Alcohol → **-ol** with lowest locant to -OH; keep the terminal **-e-** before -diol
- Ether = **alkoxyalkane** — the smaller side becomes an **-oxy** prefix on the larger chain



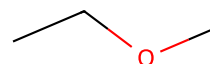
propan-1-ol

-OH → C-1



ethane-1,2-diol

keep the -e-

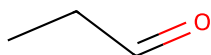


methoxyethane

alkoxy prefix

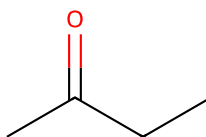
12 Aldehydes & Ketones

- Aldehyde → **-al** (-CHO carbon is always **C-1**); a -CHO on a ring → **-carbaldehyde**
- Ketone → **-one** with the lowest locant to >C=O



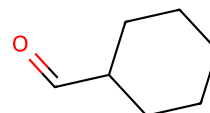
propanal

-CHO = C-1



butan-2-one

>C=O → 2

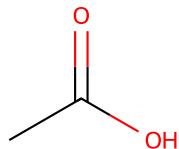


cyclohexanecarbaldehyde

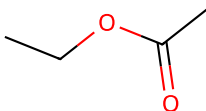
-CHO carbon not counted

13 Carboxylic Acids & Derivatives

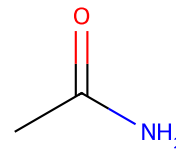
- Acid → **-oic acid** · ester → **alkyl ...-oate** · acid halide → **-oyl halide**
- Amide → **-amide** · nitrile → **-nitrile** ($\text{-C}\equiv\text{N}$ carbon is counted) · anhydride → acid name + **anhydride**



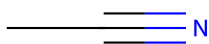
ethanoic acid
-COOH



ethyl ethanoate
alkyl ...-oate



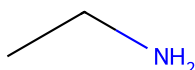
ethanamide
-amide



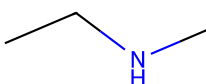
ethanenitrile
-C≡N counted

14 Amines & Nitrogen Compounds

- Primary amine → **-amine**, locant on -NH_2
- Groups on nitrogen take the italic locant **N-**. As prefixes: -NH_2 = **amino**, -NO_2 = **nitro**, $\text{-C}\equiv\text{N}$ = **cyano**



ethanamine
-NH₂ → C-1

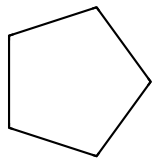


N-methylethanamine
group on N → N-

PART C · RINGS & AROMATICS

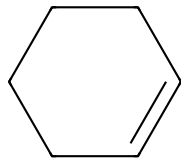
15 Cycloalkanes & Cycloalkenes

- **cyclo-** + root (C_nH_{2n}). One substituent needs **no locant** — methylcyclohexane, not “1-methyl...”
- Two or more → number for the **lowest set of locants**, ties to first-alphabetical
- Cycloalkene: the C=C carbons are fixed as **C-1 and C-2**, then number toward the nearer substituent



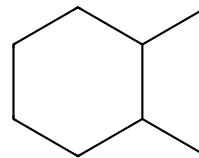
cyclopentane

5-C ring · C_5H_{10}



cyclohex-1-ene

ring C=C at 1,2



1,2-dimethylcyclohexane

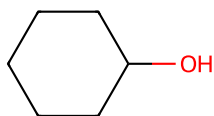
lowest set {1,2}

Ring vs chain: the parent carries the principal group. If that doesn't settle it, the **ring is chosen** (a ring outranks a chain). A ring left off the parent → **cyclohexyl-** prefix (3-cyclohexylpropanoic acid).

16 Rings Bearing a Functional Group

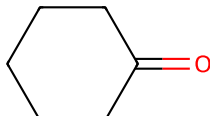
Group carbon inside the ring → ordinary suffix (counted). Group carbon outside → -carbo... suffix (not counted).

GROUP ON RING	SUFFIX	EXAMPLE (ROOT = HEX)	C COUNTED?
-OH	-ol	cyclohexan-1-ol	yes
>C=O (in ring)	-one	cyclohexanone	yes
-COOH (outside)	-carboxylic acid	cyclohexanecarboxylic acid	no
-CHO (outside)	-carbaldehyde	cyclohexanecarbaldehyde	no
-CN (outside)	-carbonitrile	cyclohexanecarbonitrile	no



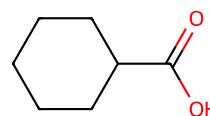
cyclohexan-1-ol

$-OH \rightarrow C-1$ · counted



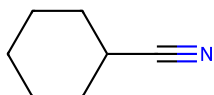
cyclohexanone

$>C=O$ in ring · counted



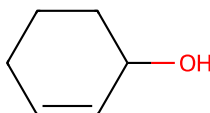
cyclohexanecarboxylic acid

C outside · not counted



cyclohexanecarbonitrile

C outside · not counted



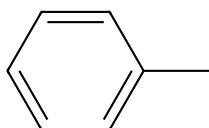
cyclohex-2-en-1-ol

$-OH$ gets $C-1$, then $C=C$

Carbon-counting trap: cyclohexanecarbaldehyde has 7 carbons in all, yet the parent root is still **hex-** — the $-CHO$ carbon sits outside the ring and is dropped from the count.

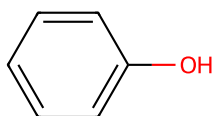
17 Benzene Derivatives

Benzene is the parent; substituents are prefixes. Retained names carry their senior group at C-1.



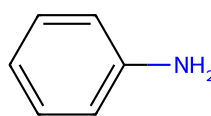
toluene

= methylbenzene



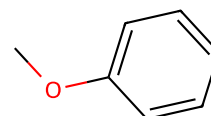
phenol

= hydroxybenzene



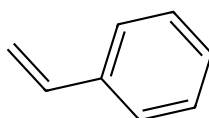
aniline

= aminobenzene



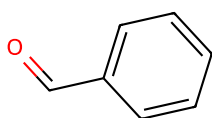
anisole

= methoxybenzene



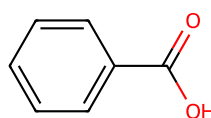
styrene

= vinylbenzene



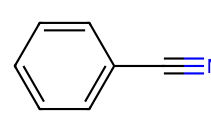
benzaldehyde

retained · group at $C-1$



benzoic acid

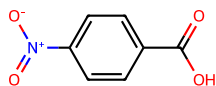
retained · group at $C-1$



benzonitrile

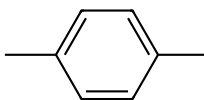
retained · group at $C-1$

Built onto retained parents · disubstituted position shorthand:



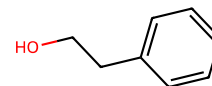
4-nitrobenzoic acid

senior -COOH at C-1



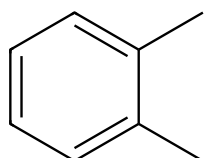
1,4-dimethylbenzene

p-xylene



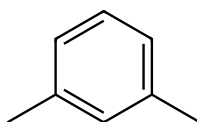
2-phenylethan-1-ol

ring = phenyl prefix



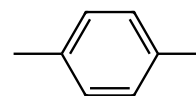
ortho (1,2)

adjacent carbons



meta (1,3)

alternate carbons



para (1,4)

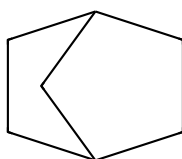
opposite carbons

phenyl vs benzene: if the principal group is on the ring, the ring is the parent (benzoic acid, phenol). If it's on a chain, the chain is the parent and the ring drops to a **phenyl-** prefix — $\text{C}_6\text{H}_5\text{-CH}_2\text{-CH}_2\text{-COOH}$ is 3-phenylpropanoic acid. No group → ring stays senior (butylbenzene).

18 Bicyclic & Spiro Systems

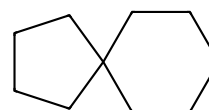
Advanced

- **Bicyclo[a.b.c]alkane:** $a \geq b \geq c$ = carbons in the three bridges *between the bridgeheads*; total C = $a+b+c+2$. Number from a bridgehead along the **longest** bridge first
- **Spiro[a.b]alkane:** one shared atom; $a \leq b$ = carbons in each ring *excluding* the spiro atom; total C = $a+b+1$. Number from the atom next to the spiro, starting in the **smaller** ring



bicyclo[2.2.1]heptane

norbornane · 7 C



spiro[4.5]decane

10 C = 4+5+1

PART D · TIE-BREAKERS & ASSEMBLY

19 Multiple Functional Groups

One suffix per name — the senior group wins it

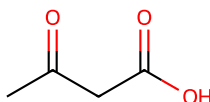
STRUCTURE	SENIOR → SUFFIX	JUNIOR → PREFIX	NAME
$\text{HO}-\text{CH}_2-\text{CH}_2-\text{NH}_2$	$-\text{OH} \rightarrow \text{-ol}$	$-\text{NH}_2 \rightarrow \text{amino}$	2-aminoethan-1-ol
$\text{OHC}-\text{CH}_2-\text{COOH}$	$-\text{COOH} \rightarrow \text{-oic acid}$	$-\text{CHO} \rightarrow \text{oxo}$	3-oxopropanoic acid
$\text{HOOC}-\text{CH}_2-\text{CO}-\text{CH}_3$	$-\text{COOH} \rightarrow \text{-oic acid}$	$>\text{C}=\text{O} \rightarrow \text{oxo}$	3-oxobutanoic acid
$\text{HO}-\text{CH}_2-\text{CH}_2-\text{CHO}$	$-\text{CHO} \rightarrow \text{-al}$	$-\text{OH} \rightarrow \text{hydroxy}$	3-hydroxypropanal
$\text{CH}_3-\text{CH}(\text{NH}_2)-\text{COOH}$	$-\text{COOH} \rightarrow \text{-oic acid}$	$-\text{NH}_2 \rightarrow \text{amino}$	2-aminopropanoic acid
$\text{CH}_3-\text{CO}-\text{CHO}$	$-\text{CHO} \rightarrow \text{-al}$	$>\text{C}=\text{O} \rightarrow \text{oxo}$	2-oxopropanal

Demoted groups become prefixes: $>\text{C}=\text{O} \rightarrow \text{oxo}$, $-\text{OH} \rightarrow \text{hydroxy}$, $-\text{NH}_2 \rightarrow \text{amino}$, $-\text{OR} \rightarrow \text{alkoxy}$. A terminal $-\text{CHO}$ written as a prefix is **oxo** (carbon in the chain); off the chain or on a ring it is **formyl**.



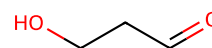
2-aminoethan-1-ol

$-\text{OH}$ suffix, $-\text{NH}_2 \rightarrow \text{amino}$



3-oxobutanoic acid

acid suffix, ketone $\rightarrow \text{oxo}$



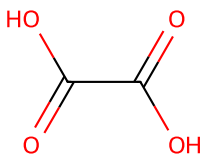
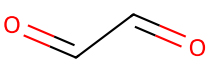
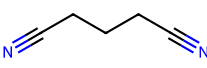
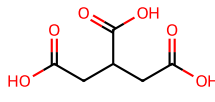
3-hydroxypropanal

$-\text{CHO}$ suffix, $-\text{OH} \rightarrow \text{hydroxy}$

20 Chain-Terminating Groups

$-COOH$, $-CHO$, $-CN$, $-COX$, $-CONH_2$ must sit at a chain end

TERMINAL GROUPS	SUFFIX FORM	COUNTED?	EXAMPLE
1 (a chain end)	-oic acid / -al / -nitrile	yes	propanoic acid · propanal · propanenitrile
2 (both ends, same)	-dioic acid / -dial / -dinitrile	yes	hexanedioic acid · ethanedial · pentanedinitrile
3+ or on a ring	-carboxylic acid / -carbaldehyde / -carbonitrile	no	propane-1,2,3-tricarboxylic acid

 ethanedioic acid 2 ends → -dioic acid	 ethanedial 2 ends → -dial	 pentanedinitrile 2 ends → -dinitrile	 propane-1,2,3-tricarboxylic acid 3 groups → -carboxylic acid
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Trap: a chain has only two ends — a third terminal-carbon group can't be -al/-oic acid/-nitrile. Switch to -carbaldehyde / -carboxylic acid / -carbonitrile and drop that carbon from the root (propane-1,2,3-tricarboxylic acid, never “propanetricarboxylic acid”). Different groups at the two ends → only the senior is the suffix (OHC-CH₂-CH₂-COOH is 4-oxobutanoic acid).

21 Alphabetical Order of Substituents

By the first point of difference in the substituent name

PREFIXES PRESENT	FILE UNDER	ORDER
4-ethyl + 2,2-dimethyl	ethyl → e, methyl → m	4-ethyl-2,2-dimethylhexane
bromo + chloro + methyl	b, c, m	1-bromo-2-chloro-4-methylbenzene
tert-butyl + ethyl + isopropyl	butyl → b, ethyl → e, isopropyl → i	butyl, ethyl, isopropyl
ethoxy + ethyl	e-t-h-o < e-t-h-y	ethoxy before ethyl

- **Ignored:** di, tri, tetra, bis, tris — “2,2-dimethyl” files under **m**; “diethyl” under **e**
- **Ignored:** italic n-, sec-, tert- → tert-butyl files under **b**
- **Counted:** iso, neo, cyclo are real letters → isopropyl under **i**, cyclohexyl under **c**
- Complex (parenthesised) → alphabetise by first letter of the whole name: (2-chloroethyl) under **c**

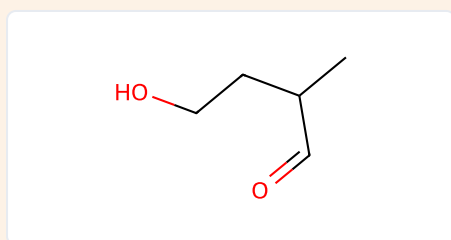
22 Writing Rules + Degree of Unsaturation

- Cite prefixes in **alphabetical order**; separate locant from word with a **hyphen** (2-methyl), numbers with **commas** (2,3-)
- Italicise: *o*-, *m*-, *p*-, *n*-, *sec*-, *tert*-, *cis*-, *trans*-, *E*-, *Z*-, *R*-, *S*-, *N*-, *O*-
- Lowest locants follow seniority: **principal group > multiple bonds > prefixes**

Degree of unsaturation $\text{DoU} = (2C + 2 + N - H - X) / 2$ — each ring or double bond = 1, each triple bond = 2. **Oxygen is not counted.**

★ · WORKED WALK-THROUGHS

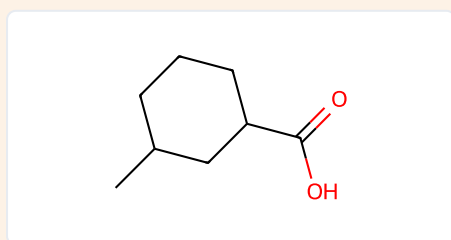
ALDEHYDE + ALCOHOL ON ONE CHAIN



- 1 –CHO outranks –OH → aldehyde is the suffix (-al), –OH drops to **hydroxy**
- 2 Parent = 4 carbons including the –CHO carbon → **butanal**
- 3 –CHO is C-1 → –OH at C-4, methyl at C-2; alphabetise hydroxy before methyl

NAME 4-hydroxy-2-methylbutanal

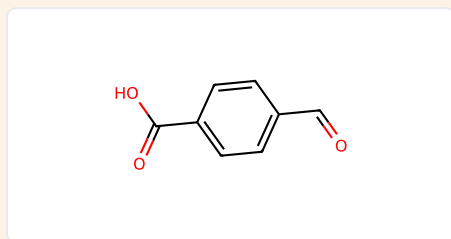
A CARBOXYLIC ACID ON A RING



- 1 Ring carries the senior group → **ring is the parent** (and a ring outranks a chain)
- 2 –COOH carbon sticks out → suffix **-carboxylic acid**, that carbon **not** counted; ring C holding it is C-1
- 3 Number to give the methyl its lowest locant → C-3

NAME 3-methylcyclohexane-1-carboxylic acid

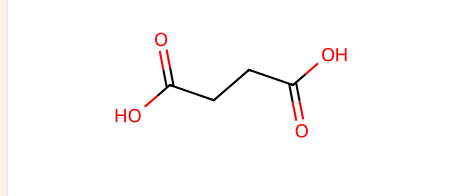
TWO GROUPS ON BENZENE



- 1 Groups: –COOH and –CHO. –COOH is senior → parent is **benzoic acid** (group at C-1)
- 2 –CHO is junior → a prefix; its carbon lies outside the ring, so **formyl** (not oxo)
- 3 Para substitution places it at C-4

NAME 4-formylbenzoic acid

A GROUP AT EACH END OF THE CHAIN



- 1 Two –COOH, one at each end → the **-dioic acid** suffix, both carboxyl carbons counted
- 2 Four carbons in the chain → butane + -dioic acid

NAME butanedioic acid

 REF · RETAINED NAMES & QUICK RECALL

COMMON NAME	IUPAC NAME
Formic acid	methanoic acid
Acetic acid	ethanoic acid
Oxalic acid	ethanedioic acid
Formaldehyde	methanal
Acetaldehyde	ethanal
Acetone	propan-2-one
Toluene	methylbenzene
Aniline	aminobenzene / benzenamine
Phenol	hydroxybenzene
Anisole	methoxybenzene
Styrene	vinylbenzene
Glycerol	propane-1,2,3-triol

Suffix priority (memorise): acid > sulphonic acid > anhydride > ester > acid halide > amide > nitrile > aldehyde > ketone > alcohol > amine. Everything junior — plus ethers, halides and nitro always — is a prefix.